

COSC/MATH 4340: Numerical Methods for Differential Equations

Numerical Linear Algebra (NLA): Linear Solvers

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Overview

Consider $A\mathbf{x} = \mathbf{b}$, or

$$\begin{bmatrix} A_{11} & \cdots & A_{1n} \\ A_{21} & \ddots & A_{2n} \\ \vdots & & \vdots \\ A_{n1} & & A_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

This has a solution $\mathbf{x} = A^{-1}\mathbf{b}$, but it is very expensive (perhaps impossible) to compute A^{-1} , which should be avoided.

Here we study different ways to find $\mathbf{x}$



Gaussian Elimination

Definition

Gaussian Elimination finds a solution using 2 steps

- ▶ Transform $A\mathbf{x} = \mathbf{b}$ into an equivalent system $U\mathbf{x} = \mathbf{g}$ that has the same solution $\mathbf{x}$, but which can be easily and efficiently solved.
- ▶ Solve $U\mathbf{x} = \mathbf{g}$ using back substitution: first solve x_n , then $x_{n-1}, x_{n-2}, \dots$

(See notes and code for example.)

Gaussian Elimination

Step 1: Implement a sequence of row operations that transforms A into U one row at a time.

- ▶ Let $A^{(1)}\mathbf{x} = \mathbf{b}^{(1)}$ denote the original equation.
- ▶ Define the row multipliers $m_{ik} = A_{ik}^{(k)} / A_{kk}^{(k)}$ for $i = k + 1, \dots, n$
- ▶ Then define a sequence of equivalent linear systems $A^{(k)}\mathbf{x} = \mathbf{b}^{(k)}$ with

$$A_{ij}^{(k+1)} = A_{ij}^{(k)} - m_{ik}A_{kj}^{(k)}, \quad i, j \in \{k + 1, \dots, n\}$$

$$b_i^{(k+1)} = b_i^{(k)} - m_{ik}b_k^{(k)}, \quad i, j \in \{k + 1, \dots, n\}$$

- ▶ After transforming all n rows, define $U = A^{(n)}$ and $\mathbf{g} = \mathbf{b}^{(n)}$.

Gaussian Elimination

Step 1 yields an equivalent system $U\mathbf{x} = \mathbf{g}$ that uses an upper triangular matrix $U = A^{(n)}$

$$\begin{bmatrix} A_{11}^{(1)} & \cdots & & & A_{1n}^{(1)} \\ 0 & A_{22}^{(2)} & & & A_{2n}^{(2)} \\ \vdots & & \ddots & & \vdots \\ 0 & \cdots & & A_{n-1,n-1}^{(n-1)} & A_{n-1,n}^{(n)} \\ 0 & \cdots & & & A_{nn}^{(n)} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} = \begin{bmatrix} b_1^{(1)} \\ b_2^{(2)} \\ \vdots \\ b_{n-1}^{(n-1)} \\ b_n^{(n)} \end{bmatrix}$$

Here, the x_j are the same, but the $a_{ij}^{(i)}$ and $b_i^{(i)}$ have changed.



Gaussian Elimination

Step 2: Solve $U\mathbf{x} = \mathbf{g}$ for the x_j using back substitution

$$x_n = g_n / U_{nn}$$

$$x_{n-1} = [g_{n-1} - U_{n-1,n}x_n] / U_{n-1,n-1}$$

$$x_{n-2} = \dots$$

$$x_k = [g_k - \sum_{j=k+1}^n U_{k,j}x_j] / U_{kk}$$

Note that you solve the x_k in reverse order: $n, n-1, n-2, \dots$



LU Decomposition

Definition

The **LU Decompositions** of a matrix A is given by $A = LU$, where $U = A^{(n)}$ is the **upper triangular** matrix resulting from Gaussian elimination and L is a **lower triangular** matrix with 1's on the diagonal and the lower-triangular entries encode the multipliers:

$$L = \begin{bmatrix} 1 & 0 & \dots & & 0 \\ m_{21} & 1 & 0 & & \vdots \\ m_{31} & m_{32} & \ddots & \ddots & \\ \vdots & \dots & \ddots & 1 & 0 \\ m_{n1} & \dots & & m_{n,n-1} & 1 \end{bmatrix}$$



LU Decomposition

Definition

The **computational complexity** of an algorithm is the maximum of the following two counts

- ▶ the total number of additions and subtractions
- ▶ the total number of multiplications and divisions

Theorem

The computational complexity for constructing an LU decomposition using Gaussian elimination is $\mathcal{O}(n^3)$

(See notes for proof.)



Error Analysis

We now study the impact of rounding errors on solutions to linear systems, starting with some background info:

Definition

A **vector norm** $||\mathbf{x}||$ is a scalar function such that

$$||\mathbf{x}|| \geq 0 \quad (1)$$

$$||\alpha\mathbf{x}|| = |\alpha| ||\mathbf{x}|| \quad (2)$$

$$||\mathbf{x} + \mathbf{y}|| \leq ||\mathbf{x}|| + ||\mathbf{y}|| \quad (3)$$

$$||\mathbf{x} - \mathbf{z}|| \leq ||\mathbf{x} - \mathbf{y}|| + ||\mathbf{y} - \mathbf{z}|| \quad (4)$$

$$| ||\mathbf{x}|| - ||\mathbf{y}|| | \leq ||\mathbf{x} - \mathbf{y}|| \quad (5)$$



Error Analysis

Definition

A **matrix norm** $\|A\|$ is a scalar function such that

$$\|A\| \geq 0 \quad (6)$$

$$\|\alpha A\| = |\alpha| \|A\| \quad (7)$$

$$\|A + B\| \leq \|A\| + \|B\| \quad (8)$$

$$\|AB\| \leq \|A\| \|B\| \quad (9)$$

$$\|A\mathbf{x}\|_{\nu} \leq \|A\| \|\mathbf{x}\|_{\nu} \quad (10)$$

where $\|\mathbf{x}\|_{\nu}$ is some vector norm.

(See notes for compatible matrix and vector norms.)



Error Analysis

The condition number measures the sensitivity of matrix-related problems to perturbations.

Definition

Let $|| \cdot ||$ denote a matrix norm, then the **condition number** $\text{cond}(A)$ of a matrix A with respect to that norm is given by:

$$\text{cond}(A) = ||A|| ||A^{-1}||.$$

Theorem

$$\text{cond}(A) \geq 1.$$

(See notes for proof.)



Error Analysis

Consider how a perturbation $\delta \mathbf{b}$ to $\mathbf{b}$ will cause a perturbation $\delta \mathbf{x}$ on the solution $\mathbf{x}$ to $A\mathbf{x} = \mathbf{b}$.

Theorem

Let $A\mathbf{x} = \mathbf{b}$ and $A(\mathbf{x} + \delta \mathbf{x}) = \mathbf{b} + \delta \mathbf{b}$. Then $A\delta \mathbf{x} = \delta \mathbf{b}$, $\delta \mathbf{x} = A^{-1}\delta \mathbf{b}$ and we have the bound

$$\frac{1}{\text{cond}(A)} \frac{\|\delta \mathbf{b}\|}{\|\mathbf{b}\|} \leq \frac{\|\delta \mathbf{x}\|}{\|\mathbf{x}\|} \leq \text{cond}(A) \frac{\|\delta \mathbf{b}\|}{\|\mathbf{b}\|}.$$

(See notes for proof.)

Error Analysis

Theorem

Consider $A\mathbf{x} = \mathbf{b}$ with nonsingular A with perturbations δA to A and $\delta \mathbf{b}$ to $\mathbf{b}$ and assume $\|\delta A\| < 1/\|A^{-1}\|$. Then $A + \delta A$ is also nonsingular and the solution to

$$(A + \delta A)(\mathbf{x} + \delta \mathbf{x}) = (\mathbf{b} + \delta \mathbf{b})$$

satisfies

$$\frac{\|\delta \mathbf{x}\|}{\|\mathbf{x}\|} \leq \frac{\text{cond}(A)}{1 - \text{cond}(A) \frac{\|\delta A\|}{\|A\|}} \left[\frac{\|\delta A\|}{\|A\|} + \frac{\|\delta \mathbf{b}\|}{\|\mathbf{b}\|} \right]$$



Error Analysis for Gaussian Elimination

Theorem

Define $\rho = \frac{1}{\|A\|_\infty} \max_{i,j,k} |A_{ij}^{(k)}|$. Then

- ▶ GE gives $LU = A + E$ and the rounding error is bounded by $\|E\|_\infty \leq n^2 \rho \|A\|_\infty \epsilon_{mach}$
- ▶ The approximated solution $\tilde{\mathbf{x}}$ using GE satisfies $(A + \delta A)\tilde{\mathbf{x}} = \mathbf{b}$ where:

$$\frac{\|\delta A\|}{\|A\|} \leq 1.01(n^3 - 3n^2)\rho\epsilon_{mach}$$

$$\frac{\|\mathbf{x} - \tilde{\mathbf{x}}\|}{\|\mathbf{x}\|} \leq \frac{\text{cond}(A)}{1 - \text{cond}(A) \frac{\|\delta A\|}{\|A\|}} \frac{\|\delta A\|}{\|A\|}.$$



Iterative Linear Solvers

Gaussian elimination (GE) is a direct solver for a linear system $A\mathbf{x} = \mathbf{b}$. That is, it is an algorithm that produces a solution $\mathbf{x}$ such that $\mathbf{x} = A^{-1}\mathbf{b}$ with some rounding error introduced. (Of course, A^{-1} is never created.)

In contrast, an iterative solver gives a sequence of approximate solutions $\mathbf{x}^{(t)}$ with the property that $\mathbf{x}^{(t)} \rightarrow \mathbf{x}$ as $t \rightarrow \infty$.

First, we will introduce some mathematical background.



Convergence Theory

Definition

A sequence of vectors $\mathbf{x}_1, \mathbf{x}_2, \dots$ **converges** to a limit $\mathbf{x} \in \mathbb{R}^n$ if and only if $\|\mathbf{x} - \mathbf{x}_k\| \rightarrow 0$ as $k \rightarrow \infty$ for some vector norm.

Definition

A sequence of matrices $A_1, A_2, \dots$ **converges** to a limit $A \in \mathbb{R}^{n \times m}$ if and only if $\|A - A_k\| \rightarrow 0$ as $k \rightarrow \infty$ for some matrix norm.

Theorem (Matrix Convergence)

Let $A \in \mathbb{R}^{n \times n}$. Then $A^t \rightarrow 0$ if and only if the spectral radius (i.e., the magnitude of the largest eigenvalue) satisfies $\rho(A) < 1$.



Convergence Theory

Theorem (Geometric series)

Let $A \in \mathbb{R}^{n \times n}$ and assume $\rho(A) < 1$. Then

$$(I - A)^{-1} = \sum_{t=0}^{\infty} A^t = I + A + A^2 + A^3 + \dots$$

(See notes.)

Theorem (Matrix Inverse Bound)

If $\|A\| < 1$ for any norm $\|\cdot\|$, then $(I - A)^{-1}$ exists and

$$\left\| (I - A)^{-1} \right\| \leq \frac{1}{1 - \|A\|}$$



Approximate Inverses

Definition

Given $A \in \mathbb{R}^{n \times n}$, C is an **approximate inverse** of A if $C \approx A^{-1}$.

Definition

The **residual matrix** of an approximate inverse is $R = I - CA$.

Theorem

Let C be an approximate inverse of A and assume $\|R\| < 1$. Then A and C are invertible and

$$\frac{\|R\|}{\|A\|\|C\|} \leq \frac{\|A^{-1} - C\|}{\|C\|} \leq \frac{\|R\|}{1 - \|R\|}$$



Approximate Inverses

Corollary

Let $\hat{\mathbf{x}} = C\mathbf{b}$ be an approximate solution to $A\mathbf{x} = \mathbf{b}$ and define the residual $\mathbf{r} = \mathbf{b} - A\hat{\mathbf{x}}$. Then

$$\|\mathbf{x} - \hat{\mathbf{x}}\| \leq \frac{\|C\mathbf{r}\|}{1 - \|R\|}$$

(See notes for proof.)



Residual Correction Method

Definition

Consider the solution $\mathbf{x}$ to $A\mathbf{x} = \mathbf{b}$ for a matrix $A \in \mathbb{R}^{n \times n}$ with an approximate inverse $C \approx A^{-1}$. Using an initial guess $\mathbf{x}^{(0)}$ to $\mathbf{x}$, the **residual correction method** constructs a sequence of estimates $\mathbf{x}^{(t)}$ defined by

$$\mathbf{x}^{(t+1)} = \mathbf{x}^{(t)} + C\mathbf{r}^{(t)},$$

where $\mathbf{r}^{(t)} = \mathbf{b} - A\mathbf{x}^{(t)}$ is the residual of estimate $\mathbf{x}^{(t)}$.

(See notes for motivation.)



Residual Correction Method

Theorem

Let $\mathbf{e}^{(t)} = \mathbf{x} - \mathbf{x}^{(t)}$ denotes the error of estimate t for the residual correction method and define $R = I - CA$. Then

$$\|\mathbf{e}^{(t)}\| \leq \|R\| \|\mathbf{e}^{(t-1)}\| \leq \|R\|^t \|\mathbf{e}^{(0)}\|$$

which linearly decays with rate $\|R\|$ if and only if $\rho(R) \leq \|R\| < 1$.

(See notes for proof.)



Residual Correction Method

Example: Consider $A(\epsilon) = A + \epsilon B$ with

$$A_0 = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 2 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 & 1 \\ -1 & 0 & 1 \\ -1 & -1 & 0 \end{bmatrix}.$$

We'd like to solve $A(\epsilon)\mathbf{x} = \mathbf{b}$ for $\mathbf{x}$ efficiently using the knowledge that when ϵ is small, then

$$A(\epsilon)^{-1} \approx C = A_0^{-1} = \begin{bmatrix} 3/4 & -1/2 & 1/4 \\ -1/2 & 1 & -1/2 \\ 1/4 & -1/2 & 3/4 \end{bmatrix}.$$

(See notes and code.)



Gauss-Jacobi Iteration

Consider each entry in $A\mathbf{x} = \mathbf{b}$:

$$b_i = \sum_{j=1}^n A_{ij}x_j = A_{ii}x_i + \sum_{j \neq i} A_{ij}x_j$$

We can solve for x_i as

$$x_i = \frac{1}{A_{ii}} \left[b_i - \sum_{j \neq i} A_{ij}x_j \right] \equiv f_i(\mathbf{x})$$

If we define $F(\mathbf{x}) = [f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_n(\mathbf{x})]^T$, then $\mathbf{x}$ must also solve

$$\mathbf{x} = F(\mathbf{x})$$



Gauss-Jacobi Iteration

Definition

For some initial guess $\mathbf{x}^{(0)} \in \mathbb{R}^n$, **Gauss-Jacobi Iteration** is defined by

$$x_i^{(t+1)} = \frac{1}{A_{ii}} \left[b_i - \sum_{j \neq i} A_{ij} x_j^{(t)} \right].$$



Gauss-Jacobi Iteration

Theorem

Let

$$\mu = \max_i \sum_{j \neq i} \frac{|A_{ij}|}{|A_{ii}|}$$

then the error $\mathbf{e}^{(t)} = \mathbf{x} - \mathbf{x}^{(t)}$ for Gauss-Jacobi iteration satisfies

$$\|\mathbf{e}^{(t+1)}\|_{\infty} \leq \mu \|\mathbf{e}^{(t)}\|_{\infty} \leq \mu^{t+1} \|\mathbf{e}^{(0)}\|_{\infty}.$$

and $\|\mathbf{e}^{(t)}\|_{\infty} \rightarrow 0$ if $\mu < 1$.

(See notes and code.)



Gauss-Jacobi Iteration

Definition

A matrix A is **diagonally dominant** if for each i ,

$$\sum_{j \neq i} |A_{ij}| < |A_{ii}|$$

Theorem

Gauss-Jacobi iteration converges for any diagonally dominant matrix.

Proof: $\mu < 1$ since $\sum_{j \neq i} \frac{|A_{ij}|}{|A_{ii}|} < 1$ for every i .

(See notes for example.)



Gauss-Seidel Iteration

Consider each entry in $A\mathbf{x} = \mathbf{b}$:

$$b_i = \sum_{j=1}^n A_{ij}x_j = \sum_{j=1}^{i-1} A_{ij}x_j + A_{ii}x_i + \sum_{j=i+1}^n A_{ij}x_j$$

We can solve for x_i as

$$x_i = \frac{1}{A_{ii}} \left[b_i - \sum_{j=1}^{i-1} A_{ij}x_j - \sum_{j=i+1}^n A_{ij}x_j \right].$$

When computing $x_i^{(t+1)}$ with Gauss Seidel, we might as well use the values $x_j^{(t+1)}$ for $j \leq i$ since they are already computed.



Gauss-Seidel Iteration

Definition

For some initial guess $\mathbf{x}^{(0)} \in \mathbb{R}^n$, **Gauss-Seidel Iteration** is defined by

$$x_i^{(t+1)} = \frac{1}{A_{ii}} \left[b_i - \sum_{j=1}^{i-1} A_{ij} x_j^{(t+1)} - \sum_{j=i+1}^n A_{ij} x_j^{(t)} \right].$$

Note that this is an *implicit method* because estimates $x_i^{(t+1)}$ depend on other $x_j^{(t+1)}$ values at iteration $t + 1$.



Gauss-Seidel Iteration

Theorem

The error $\mathbf{e}^{(t)} = \mathbf{x} - \mathbf{x}^{(t)}$ for Gauss-Seidel iteration satisfies

$$\|\mathbf{e}^{(t+1)}\|_{\infty} \leq \eta \|\mathbf{e}^{(t)}\|_{\infty} \leq \eta^{t+1} \|\mathbf{e}^{(0)}\|_{\infty},$$

where $\eta = \max_i \frac{\beta_i}{1-\alpha_i}$ and

$$\alpha_i = \sum_{j=1}^{i-1} \frac{|A_{ij}|}{|A_{ii}|} \text{ and } \beta_i = \sum_{j=i+1}^n \frac{|A_{ij}|}{|A_{ii}|}$$

with $\alpha_i + \beta_i < 1$. Note that $\|\mathbf{e}^{(t)}\|_{\infty} \rightarrow 0$ iff $\eta < 1$.

Gauss-Seidel has faster convergence than Gauss-Jacobi: $\eta \leq \mu < 1$.



General Iterative Methods

To solve $A\mathbf{x} = \mathbf{b}$, consider the split A into two matrices:
 $A = N + P$. The goal is to choose some N so that it is easy to solve $N\mathbf{x} = \mathbf{b}$. Then one can just define $P = A - N$.

It follows that $(N + P)\mathbf{x} = \mathbf{b}$, and

$$N\mathbf{x} = \mathbf{b} - P\mathbf{x}$$

One can then iteratively obtain $\mathbf{x}^{(t+1)}$ using $\mathbf{x}^{(t)}$ by solving

$$N\mathbf{x}^{(t+1)} = \mathbf{b} - P\mathbf{x}^{(t)}.$$

(See notes for examples.)



General Iterative Methods

Theorem

Let $M = N^{-1}P$. Then the error $\mathbf{e}^{(t)} = \mathbf{x} - \mathbf{x}^{(t)}$ for a general iterative method satisfies

$$\|\mathbf{e}^{(t+1)}\| \leq \|M\| \|\mathbf{e}^{(t)}\| \leq \|M\|^{t+1} \|\mathbf{e}^{(0)}\|.$$

and $\|\mathbf{e}^{(t)}\| \rightarrow 0$ iff $\|M\| < 1$ (which is the same as $\rho(M) < 1$).

(See notes.)



General Iterative Methods

Considering $A = N + P$

- ▶ Gauss-Jacobi iteration and Gauss-Seidel iteration are just two particular choices for N and P .
- ▶ They can be implemented and their error analyzed using this more general framework.
- ▶ For Gauss-Jacobi, one can show that $\mu < 1$ implies $\|M\| < 1$
- ▶ For Gauss-Seidel, one can show that $\eta < 1$ implies $\|M\| < 1$



General Iterative Methods

Gauss-Jacobi corresponds to when

$$N = \begin{bmatrix} A_{11} & 0 & \dots & 0 \\ 0 & A_{22} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \dots & 0 & A_{nn} \end{bmatrix}, \quad P = \begin{bmatrix} 0 & A_{12} & \dots & A_{1n} \\ A_{21} & 0 & \ddots & \vdots \\ \vdots & \ddots & \ddots & A_{n-1,n} \\ A_{n1} & \dots & A_{n,n-1} & 0 \end{bmatrix}$$

In that case we iteratively solve

$$\mathbf{x}^{(t+1)} = N^{-1}[\mathbf{b} - P\mathbf{x}^{(t)}]$$

which uses knowledge that $N^{-1} = \text{diag}[A_{11}^{-1}, A_{22}^{-1}, \dots, A_{nn}^{-1}]$.



General Iterative Method

Gauss-Seidel corresponds to when

$$N = \begin{bmatrix} A_{11} & 0 & \dots & 0 \\ A_{21} & A_{22} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ A_{n,1} & \dots & A_{n,n-1} & A_{nn} \end{bmatrix}, \quad P = \begin{bmatrix} 0 & A_{12} & \dots & A_{1n} \\ 0 & 0 & \ddots & \vdots \\ \vdots & \ddots & \ddots & A_{n-1,n} \\ 0 & \dots & 0 & 0 \end{bmatrix}$$

In that case we iteratively solve

$$N\mathbf{x}^{(t+1)} = [\mathbf{b} - P\mathbf{x}^{(t)}]$$

Since N is a lower triangular matrix, we can easily solve the linear system.

Sparse Matrices

Matrices often contain mostly zeros (specifically $\mathcal{O}(N)$ zeros) and are called *sparse*. Such matrices are efficiently encoded by keeping track of where the nonzero entries are what their values are.

$$\begin{pmatrix} 1.0 & 0 & 5.0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 3.0 & 0 & 0 & 0 & 0 & 11.0 & 0 \\ 0 & 0 & 0 & 0 & 9.0 & 0 & 0 & 0 \\ 0 & 0 & 6.0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 7.0 & 0 & 0 & 0 & 0 \\ 2.0 & 0 & 0 & 0 & 0 & 10.0 & 0 & 0 \\ 0 & 0 & 0 & 8.0 & 0 & 0 & 0 & 0 \\ 0 & 4.0 & 0 & 0 & 0 & 0 & 0 & 12.0 \end{pmatrix}$$

(See notes for example.)



Sparse Matrix Operations

Theorem

Let $\mathbf{x}$ be a length- n vector and let A be a matrix of size $n \times n$. Let m denote the number of nonzero entries in A . Then the computational complexity of matrix-vector multiplication, $A\mathbf{x}$, is

- ▶ $\mathcal{O}(n^2)$ if the matrix is encoded as a 'dense' matrix; and
- ▶ $\mathcal{O}(m)$ if the matrix is encoded as a 'sparse' matrix.

(See notes and code.)



Sparse Matrix Operations

Sparse matrix encodings allow one to study massive matrices containing millions of rows and columns.

In those cases, often 99.999% of the matrix entries are zeros and one must develop numerical linear algebra methods that allow the matrices to stay sparse.

- ▶ Gaussian elimination is not good for sparse matrices, since the row operations can convert a sparse matrix into a dense matrix.
- ▶ The iterative methods(residual correction, Gauss-Jacobi and Gauss-Seidel) can benefit from sparse matrices.
- ▶ SciPy is a Python module includes ways to 'sparsify' Numpy arrays and solve sparse linear systems.